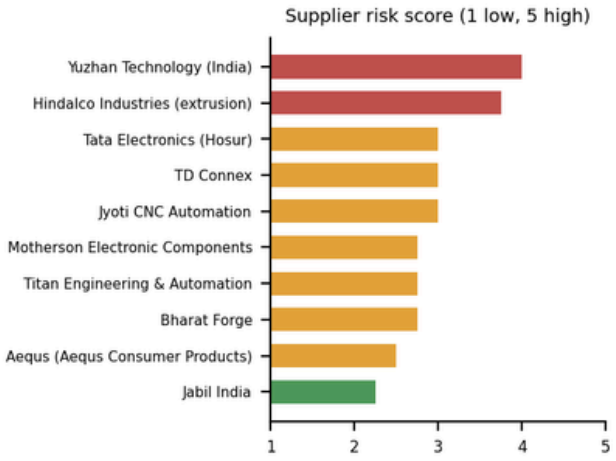
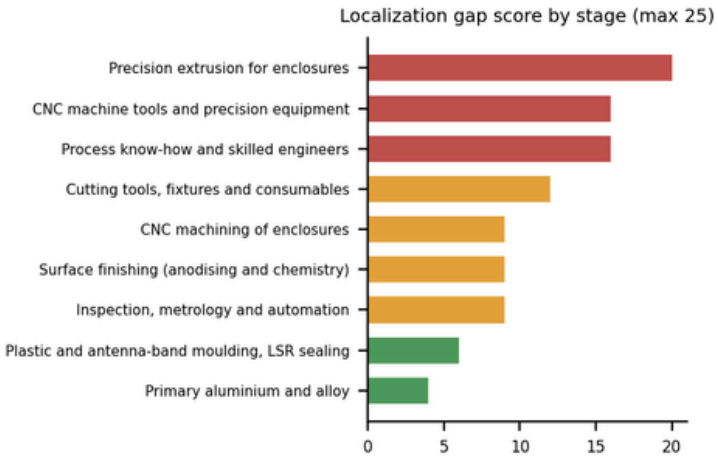
Localization gap and supplier risk: mobile aluminium enclosures in India

What the map shows

Indiacan alreadymachine enclosures; the gaps sit upstream and underneath. 3 of 9 value-chain stages are Critical: precision extrusion for enclosures (gap 20/25); CNC machine tools and precision equipment (gap 16/25); process know-how and skilled engineers (gap 16/25). Phone-enclosure aluminium extrusion is still 100% imported until Hindalco's Kuppam plant starts (target March 2027).

Of 10 suppliers scored, 2 rate High risk: Yuzhan Technology (India) (4.00) and Hindalco Industries (extrusion) (3.75). The average score is 2.98 of 5. Risk is driven less by any one company than by shared dependencies: imported extrusion, China-linked equipment and know-how, and several new plants ramping at once.





Top three risks and what they are worth

Rank	Risk	Score now	Score after mitigation	Reduction
1	R1: Imported aluminium extrusion dependence	20	9	55%
2	R2: Chinese equipment and engineer dependence	20	12	40%
3	R3: Capacity ramp and execution delay	16	9	44%

Link to the TCO model (Project 1): at today's duty, freight and incentive assumptions a local Indian enclosure lands about 9.6% above a China import and needs roughly 95% first-pass yield to match it. That makes yield and quality ramp (R5, score 12) the gating factor for shifting volume, and it is why the plan releases volume on milestones rather than on announcements.

Mitigation actions for the top three risks

Ownersarefunctions,notnamedpeople.KPIsaretheearly-warningsignals to review weekly or monthly.

Risk	#	Action	Owner	Timing	KPI / trigger
R1: Imported aluminium extrusion dependence	1	Stage-gate qualification of Hindalco Kuppam extrusion: sample lots, dimensional and surface capability studies, then volume release tied to measured yield, ahead of the March 2027 start.	Sourcing + Supplier quality	0-9 months	Qualified lots passed; domestic extrusion share of total
	2	Hold safety stock and keep two approved import sources for extrusion until domestic yield is proven; set weeks of cover by lead time and lane risk.	Supply chain planning	0-3 months	Weeks of extrusion cover
	3	Move extrusion pricing to an LME-linked index with a collar so metal price risk is shared and not buried in unit price.	Sourcing (commodity team)	3-6 months	Share of volume on index pricing
	4	Pre-plan duty and customs treatment (bonded storage, duty-exemption eligibility) for imported extrusion and review quarterly against policy changes.	Sourcing + Trade compliance	0-6 months	Duty paid versus plan
R2: Chinese equipment and engineer dependence	1	Run side-by-side capability studies of Indian machine and tooling makers (Titan Engineering, Jyoti CNC, Bharat Forge) against incumbent equipment on one pilot line.	Engineering + Supplier development	0-9 months	Pilot-line OEE and tolerance capability versus incumbent
	2	Dual-source critical equipment from non-China origins (for example Japan, Taiwan, Korea, Germany) and cut China's share of new equipment orders.	Sourcing + Engineering	3-12 months	Share of new equipment value from non-China origin
	3	Build a skilling and tool-maintenance programme in India (train and certify local technicians and process engineers) so tool upkeep no longer depends on imported engineers.	Supplier development + HR partners	0-12 months	Certified Indian process engineers; mean time to repair
	4	Stock critical spares and consumable tooling in India with a defined buffer, and agree emergency supply terms with equipment makers.	Supply chain planning	0-3 months	Days of critical-spares cover
R3: Capacity ramp and execution delay	1	Release volume to each new plant (Kanchipuram sites, Hosur, Kuppam) only on milestone gates for capacity, yield and quality.	Sourcing + Operations	0-12 months	Gate passed on schedule
	2	Keep parallel supply across Tata, Yuzhan, Motherson and Aequs so one slip does not stop launch volume; cap any single plant's share.	Sourcing (commodity team)	0-6 months	Largest single-plant share of volume
	3	Run a weekly ramp review (schedule, OEE, first-pass yield) with each supplier and escalate any slip beyond an agreed number of weeks.	Operations + Supplier quality	0-12 months	Schedule slip in weeks; first-pass yield versus plan
	4	Keep contingency capacity at existing China or Vietnam enclosure lines during the first launch cycle of each new Indian line.	Supply chain planning	0-12 months	Contingency capacity available

Sequencing: first 90 days

- **Weeks 0-4:** setweeks-of-cover targets for extrusion and critical spares; confirm second import source for extrusion (R1, R2).
- **Weeks 4-8:** agree milestone gates and weekly ramp reviews with each new-plant supplier; define the single-plant share ceiling (R3).
- **Weeks 8-12:** start the equipment capability pilot with Indian machine makers and launch the technician certification plan (R2); open Kuppam sample-lot planning (R1).

What this does not cover

Supplier and stagescores aremy judgments on public press reports, some based on unnamed sources, and plant status changes quickly. 3 of 9 value-chain stages (cutting tools, surface finishing, inspection) have no direct public source and are marked Analyst estimate: verify them with suppliers before use. Every score is an editable input in the workbook, so the ranking updates if you disagree.

Sources are listed on the Sources sheet of the workbook (ECMS approval notices, Business Standard, Digitimes, Bloomberg via Deccan Herald, Rest of World, Reuters).